

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano history.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc history.c -o history
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./history
Command History Demo
Type 'history' to display history
Type 'exit' to quit

$ ls
$ pwd
$ date
$ whoami
$ history

--- Command History ---
1  ls
2  pwd
3  date
4  whoami
$ ^Z
[1]+  Stopped                  ./history
sai_pranavi@LAPTOP-RVOHSPKQ:~$
```

Code:



sai_pranavi@LAPTOP-RVOHSI



GNU nano 8.7.1

history.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_HISTORY 5
#define MAX_CMD 100

char history[MAX_HISTORY][MAX_CMD];
int count = 0;

void add_history(char *cmd)
{
    if (count < MAX_HISTORY)
    {
        strcpy(history[count], cmd);
        count++;
    }
    else
    {
        for (int i = 1; i < MAX_HISTORY; i++)
            strcpy(history[i - 1], history[i]);

        strcpy(history[MAX_HISTORY - 1], cmd);
    }
}

void display_history()
{
    printf("\n--- Command History ---\n");

    for (int i = 0; i < count; i++)
        printf("%d %s\n", i + 1, history[i]);
}

int main()
{
    char command[MAX_CMD];

    printf("Command History Demo\n");
    printf("Type 'history' to display history\n");
    printf("Type 'exit' to quit\n\n");

    while (1)
    {
        printf("$ ");
        fgets(command, MAX_CMD, stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
            break;

        if (strcmp(command, "history") == 0)
        {
            display_history();
            continue;
        }

        if (strlen(command) > 0)
            add_history(command);
    }

    return 0;
}
```

11B.

```
pid_t p2 = fork();

if (p2 == -1)
{
    perror("fork");
    return 1;
}

if (p2 == 0)
{
    // Second command: wc -l
    close(pipefd[1]);

    dup2(pipefd[0], STDIN_FILENO);
    close(pipefd[0]);

    execlp("wc", "wc", "-l", NULL);

    perror("execlp");
    exit(1);
}

// Parent
close(pipefd[0]);
close(pipefd[1]);

waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);

return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano pipeline.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc pipeline.c -o pipeline
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./pipeline
203
sai_pranavi@LAPTOP-RVOHSPKQ:~$ |
```

12A

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int fd[2];

    if (pipe(fd) == -1)
    {
        perror("pipe");
        return 1;
    }

    pid_t pid = fork();

    if (pid == -1)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        // Child: execute "ls"
        close(fd[0]);

        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);

        execlp("ls", "ls", NULL);

        perror("execlp");
        exit(1);
    }
    else
    {
        // Parent: execute "wc -l"
        close(fd[1]);

        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);

        execlp("wc", "wc", "-l", NULL);

        perror("execlp");
        wait(NULL);
    }

    return 0;
}

```

```

sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano pipe_basic.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc pipe_basic.c -o pipe_basic
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./pipe_basic
205
sai_pranavi@LAPTOP-RVOHSPKQ:~$ |

```

12B

```
sai_pranavi@LAPTOP-RVOHSI × + ▾
GNU nano 8.7.1 multi_pipe.c *

// Process 3: wc -l
pid_t p3 = fork();

if (p3 == 0)
{
    dup2(pipe2[0], STDIN_FILENO);

    close(pipe1[0]);
    close(pipe1[1]);
    close(pipe2[0]);
    close(pipe2[1]);

    execlp("wc", "wc", "-l", NULL);

    perror("wc");
    exit(1);
}

// Parent closes all pipe descriptors
close(pipe1[0]);
close(pipe1[1]);
close(pipe2[0]);
close(pipe2[1]);

// Wait for all processes
waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);
waitpid(p3, NULL, 0);

return 0;
}|
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano multi_pipe.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc multi_pipe.c -o multi_pipe
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./multi_pipe
116
sai_pranavi@LAPTOP-RVOHSPKQ:~$ |
```

13A

```
sai_pranavi@LAPTOP-RVOHSI × + ▾
GNU nano 8.7.1 large_pipeline.c *
    close(p3[0]); close(p3[1]);
    close(p4[0]); close(p4[1]);

    execlp("wc", "wc", "-l", NULL);

    perror("wc");
    exit(1);
}

// Parent closes all descriptors
close(p1[0]); close(p1[1]);
close(p2[0]); close(p2[1]);
close(p3[0]); close(p3[1]);
close(p4[0]); close(p4[1]);

// Wait for all processes
waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);
waitpid(pid3, NULL, 0);
waitpid(pid4, NULL, 0);
waitpid(pid5, NULL, 0);

clock_t end = clock();

double time_taken =
    (double)(end - start) / CLOCKS_PER_SEC;

printf("\nPipeline completed successfully.\n");
printf("Execution time: %.6f seconds\n", time_taken);

return 0;
}|
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano large_pipeline.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc large_pipeline.c -o large_pipeline
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./large_pipeline
117

Pipeline completed successfully.
Execution time: 0.001630 seconds
```



sai_pranavi@LAPTOP-RVOHSI



```
Pipeline completed successfully.
Execution time: 0.001630 seconds
sai_pranavi@LAPTOP-RVOHSI:~$ cd ~/ShellForge
-bash: cd: /home/sai_pranavi/ShellForge: No such file or directory
sai_pranavi@LAPTOP-RVOHSI:~$ cd ~/skill
sai_pranavi@LAPTOP-RVOHSI:~/skill$ ls
simplecommandloop  simplecommandloop.c
sai_pranavi@LAPTOP-RVOHSI:~/skill$ cd -
mkdir -p ShellForge/src ShellForge/tests ShellForge/docs
sai_pranavi@LAPTOP-RVOHSI:~$ cp ~/skill/* ShellForge/src/
sai_pranavi@LAPTOP-RVOHSI:~$ cd ~/ShellForge
ls src
simplecommandloop  simplecommandloop.c
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ nano Makefile
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ make
gcc -Wall -Wextra -g src/simplecommandloop.c -o shellforge
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ ./shellforge
2520030514_SHELLFORGE$?Z
[2]+  Stopped                  ./shellforge
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ nano README.md
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ ./shellforge
2520030514_SHELLFORGE$?Z
[3]+  Stopped                  ./shellforge
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ find . -maxdepth 2 -type f
./src/simplecommandloop.c
./src/simplecommandloop
./Makefile
./README.md
./shellforge
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to 'main' in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/sai_pranavi/ShellForge/.git/
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git add .
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git commit -m "Complete ShellForge project"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: empty ident name (for <sai_pranavi@LAPTOP-RVOHSI.localdomain>) not allowed
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git tag v1.0
fatal: Failed to resolve 'HEAD' as a valid ref.
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git config --global user.name "Sai Pranavi"
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git config --global user.email "YOUR_GITHUB_EMAIL"
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git config --global user.name
git config --global user.email
Sai Pranavi
YOUR_GITHUB_EMAIL
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git config --global user.email "saipranavi@gmail.com"
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git config --global user.name
git config --global user.email
Sai Pranavi
saipranavi@gmail.com
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git config --global user.email "pranavinaliveni05@gmail.com"
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git config --global user.name
git config --global user.email
Sai Pranavi
pranavinaliveni05@gmail.com
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git add .
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git commit -m "Complete ShellForge project"
[master (root-commit) b647125] Complete ShellForge project
 5 files changed, 54 insertions(+)
 create mode 100644 Makefile
 create mode 100644 README.md
 create mode 100755 shellforge
 create mode 100755 src/simplecommandloop
 create mode 100644 src/simplecommandloop.c
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git tag
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git tag v1.0
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git tag
v1.0
sai_pranavi@LAPTOP-RVOHSI:~/ShellForge$ git status
On branch master
nothing to commit, working tree clean
```



sai_pranavi@LAPTOP-RVOHSI



```
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ git tag v1.0
git status
fatal: tag 'v1.0' already exists
On branch master
nothing to commit, working tree clean
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ ./shellforge
2520030514_SHELLFORGE$ls
you entered ls
2520030514_SHELLFORGE$pwd
you entered pwd
2520030514_SHELLFORGE$echo Hello
you entered echo Hello
2520030514_SHELLFORGE$ls | wc -l
you entered ls | wc -l
2520030514_SHELLFORGE$^C
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ cd ~/ShellForge
cp ~/skill/* src/
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ ls src
simplecommandloop simplecommandloop.c
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ make clean
make
rm -f shellforge
gcc -Wall -Wextra -g src/simplecommandloop.c -o shellforge
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ ls -l
total 40
-rw-r--r-- 1 sai_pranavi sai_pranavi 147 Sep 29 18:03 Makefile
-rw-r--r-- 1 sai_pranavi sai_pranavi 337 Sep 29 18:03 README.md
drwxr-xr-x 2 sai_pranavi sai_pranavi 4096 Sep 29 18:02 docs
-rwxr-xr-x 1 sai_pranavi sai_pranavi 19024 Sep 29 18:10 shellforge
drwxr-xr-x 2 sai_pranavi sai_pranavi 4096 Sep 29 18:02 src
drwxr-xr-x 2 sai_pranavi sai_pranavi 4096 Sep 29 18:02 tests
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ nano demo.sh
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ chmod +x demo.sh
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ ./demo.sh
===== SHELLFORGE FINAL DEMO =====

1. Simple Command
=====
Makefile README.md demo.sh docs shellforge src tests

2. Working Directory
=====
/home/sai_pranavi/ShellForge

3. Pipeline
=====
7

4. Multiple Pipeline
=====
2

5. Output Redirection
=====
ShellForge Test

6. Background Job
=====
Background job started

===== DEMO COMPLETED =====
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    demo.sh
    demo.txt

nothing added to commit but untracked files present (use "git add" to track)
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ git add demo.sh demo.txt
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ git commit -m "Add final demo scenarios and validation"
[master d687a2f] Add final demo scenarios and validation
2 files changed, 39 insertions(+)
create mode 100755 demo.sh
create mode 100644 demo.txt
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ git tag v1.1
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ git tag
v1.0
v1.1
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ git status
On branch master
nothing to commit, working tree clean
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$ git log --oneline --decorate --all
d687a2f (HEAD -> master, tag: v1.1) Add final demo scenarios and validation
b647125 (tag: v1.0) Complete ShellForge project
sai_pranavi@LAPTOP-RVOHSPHQ:~/ShellForge$
```




sai_pranavi@LAPTOP-RVOHSI



```
[VALID] ls | grep .c > result.txt
```

```
=====
Q15 TESTS COMPLETED
=====
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ ls -l output.txt error.txt merged.txt
```

```
-rw-r--r-- 1 sai_pranavi sai_pranavi 25 Sep 29 18:24 error.txt
```

```
-rw-r--r-- 1 sai_pranavi sai_pranavi 30 Sep 29 18:24 merged.txt
```

```
-rw-r--r-- 1 sai_pranavi sai_pranavi 29 Sep 29 18:24 output.txt
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ echo "----- output.txt -----"
```

```
cat output.txt
```

```
echo "----- error.txt -----"
```

```
cat error.txt
```

```
echo "----- merged.txt -----"
```

```
cat merged.txt
```

```
----- output.txt -----
```

```
Hello from redirected stdout
```

```
----- error.txt -----
```

```
This is an error message
```

```
----- merged.txt -----
```

```
stdout message
```

```
stderr message
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ grep -n "dup2" src/complex_shell.c
```

```
33:         dup2(fd, STDOUT_FILENO);
```

```
63:         dup2(fd, STDERR_FILENO);
```

```
93:         dup2(fd, STDOUT_FILENO);
```

```
94:         dup2(STDOUT_FILENO, STDERR_FILENO);
```

```
127:         dup2(pipefd[1], STDOUT_FILENO);
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ git status
```

```
On branch master
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
```

```
complex_shell
```

```
error.txt
```

```
merged.txt
```

```
output.txt
```

```
src/complex_shell.c
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ mv complex_shell src/
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ mkdir -p tests
```

```
mv output.txt error.txt merged.txt tests/
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ find . -maxdepth 2 -type f | sort
```

```
./.git/COMMIT_EDITMSG
```

```
./.git/HEAD
```

```
./.git/config
```

```
./.git/description
```

```
./.git/index
```

```
./.gitignore
```

```
./Makefile
```

```
./README.md
```

```
./demo.sh
```

```
./demo.txt
```

```
./shellforge
```

```
./src/complex_shell
```

```
./src/complex_shell.c
```

```
./src/simplecommandloop
```

```
./src/simplecommandloop.c
```

```
./tests/error.txt
```

```
./tests/merged.txt
```

```
./tests/output.txt
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ git status
```

```
On branch master
```

```
Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
```

```
src/complex_shell
```

```
src/complex_shell.c
```

```
tests/
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ git add src/complex_shell.c src/complex_shell tests/
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ git commit -m "Add complex redirection and pipeline integration"
```

```
[master 59bfaef] Add complex redirection and pipeline integration
```

```
5 files changed, 214 insertions(+)
```

```
create mode 100755 src/complex_shell
```

```
create mode 100644 src/complex_shell.c
```

```
create mode 100644 tests/error.txt
```

```
create mode 100644 tests/merged.txt
```

```
create mode 100644 tests/output.txt
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ git tag v1.2
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ git tag
```

```
v1.0
```

```
v1.1
```

```
v1.2
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ git status
```

```
On branch master
```

```
nothing to commit, working tree clean
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$ git log --oneline --decorate --all
```

```
59bfaef (HEAD -> master, tag: v1.2) Add complex redirection and pipeline integration
```

```
d687a2f (tag: v1.1) Add final demo scenarios and validation
```

```
b647125 (tag: v1.0) Complete ShellForge project
```

```
sai_pranavi@LAPTOP-RVOHSPWQ:~/ShellForge$
```

```

#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

#define MAX_JOBS 20

typedef struct
{
    int job_id;
    pid_t pid;
    pid_t pgid;
    char command[100];
    char state[20];
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;
int next_job_id = 1;

void add_job(pid_t pid, pid_t pgid, const char *command)
{
    if (job_count >= MAX_JOBS)
    {
        printf("Job table full.\n");
        return;
    }

    jobs[job_count].job_id = next_job_id++;
    jobs[job_count].pid = pid;
    jobs[job_count].pgid = pgid;

    strncpy(jobs[job_count].command,
            command,
            sizeof(jobs[job_count].command) - 1);

    jobs[job_count].command[
        sizeof(jobs[job_count].command) - 1
    ] = '\0';

    strcpy(jobs[job_count].state, "RUNNING");

    printf("[%d] %d started in background: %s\n",
        jobs[job_count].job_id,
        pid,
        command);

    job_count++;
}

void remove_job(int index)
{
    for (int i = index; i < job_count - 1; i++)
    {
        jobs[i] = jobs[i + 1];
    }

    job_count--;
}

void update_jobs()
{
    int status;

    for (int i = 0; i < job_count; i++)
    {
        pid_t result = waitpid(
            jobs[i].pid,
            &status,
            WNOHANG
        );

        if (result == 0)
        {
            i++;
        }
        else if (result == jobs[i].pid)
        {
            strcpy(jobs[i].state, "DONE");

            printf("\n[%d] DONE: %s\n",
                jobs[i].job_id,
                jobs[i].command);

            remove_job(i);
        }
        else
        {

```

```
sai_pranavi@LAPTOP-RVOHSI x + v
sai_pranavi@LAPTOP-RVOHSPKQ:~$ cd ~/ShellForge
nano src/job_control.c
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ gcc -Wall -Wextra -g src/job_control.c -o src/job_control
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ ./src/job_control
=====
SHELLFORGE JOB CONTROL
=====

Commands:
bg      - start background job
jobs    - list active jobs
check   - update job states
exit    - quit

SHELLFORGE-JOBS$ bg
[1] 639 started in background: background-task

SHELLFORGE-JOBS$ Background process 639 running for 10 seconds...
Background process 639 finished.

[1] DONE: background-task

SHELLFORGE-JOBS$ exit
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ |
```

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```
GNU nano 8.7.1 activejobs.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid1, pid2;

    pid1 = fork();

    if (pid1 == 0) {
        sleep(5);
        return 0;
    }

    pid2 = fork();

    if (pid2 == 0) {
        sleep(3);
        return 0;
    }

    printf("Active Jobs:\n");
    printf("Job 1 PID: %d\n", pid1);
    printf("Job 2 PID: %d\n", pid2);

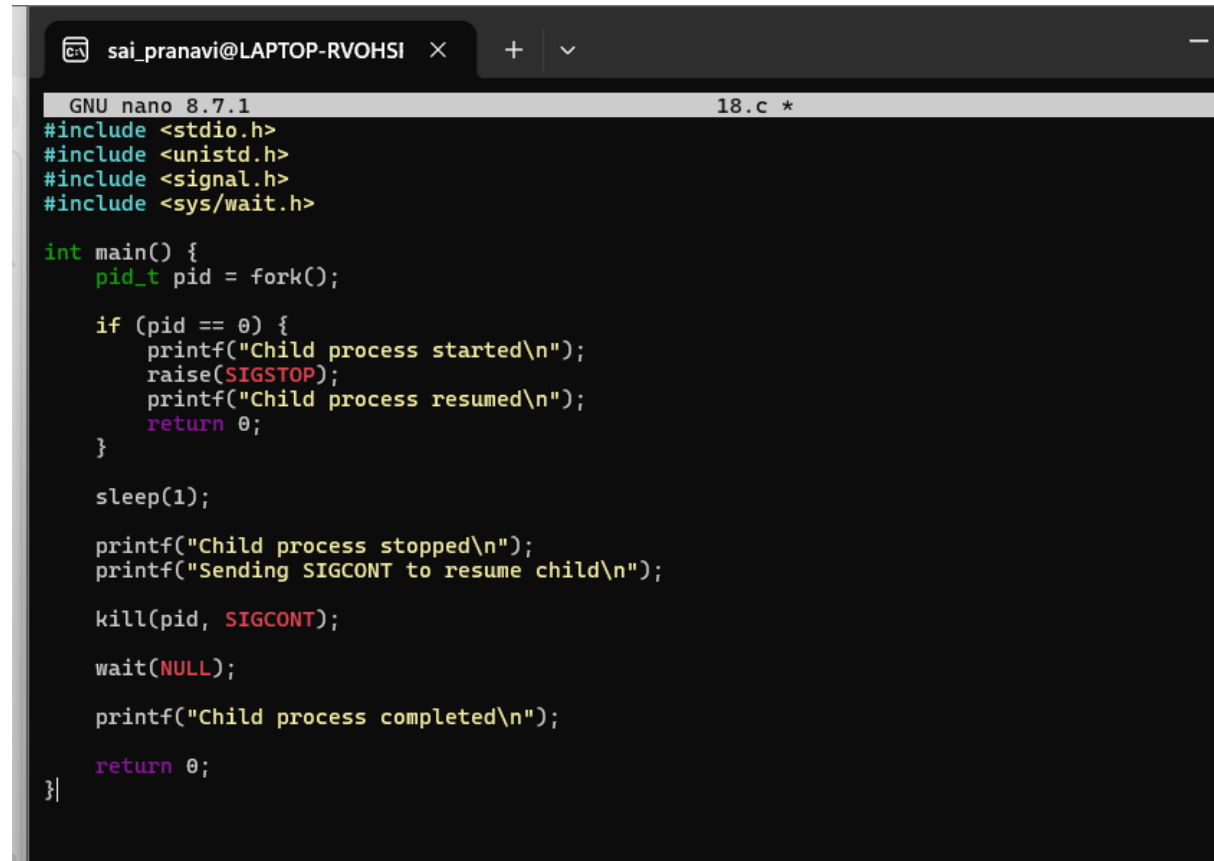
    wait(NULL);
    wait(NULL);

    printf("All jobs completed\n");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ nano activejobs.c
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ gcc activejobs.c -o activejobs
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ ./activejobs
Active Jobs:
Job 1 PID: 1228
Job 2 PID: 1229
All jobs completed
```

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```
GNU nano 8.7.1 18.c *
#include <stdio.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork();

    if (pid == 0) {
        printf("Child process started\n");
        raise(SIGSTOP);
        printf("Child process resumed\n");
        return 0;
    }

    sleep(1);

    printf("Child process stopped\n");
    printf("Sending SIGCONT to resume child\n");

    kill(pid, SIGCONT);

    wait(NULL);

    printf("Child process completed\n");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ gcc q18.c -o q18
cc1: fatal error: q18.c: No such file or directory
compilation terminated.
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ gcc 18.c -o 18
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ ./18
Child process started
Child process stopped
Sending SIGCONT to resume child
Child process resumed
Child process completed
```

```

GNU nano 8.7.1 19.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_sigint(int sig) {
    printf("\nSIGINT received\n");
    printf("Interrupt handled safely\n");
}

int main() {
    signal(SIGINT, handle_sigint);

    printf("Program running...\n");
    printf("Press Ctrl+C to send SIGINT\n");

    while (1) {
        sleep(1);
    }

    return 0;
}

```

```

sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ nano 19.c
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ gcc 19.c -o 19
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ ./19
Program running...
Press Ctrl+C to send SIGINT
^C
SIGINT received
Interrupt handled safely

^C
SIGINT received
Interrupt handled safely
^Z
[1]+  Stopped                  ./19
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ |

```

```

sai_pranavi@LAPTOP-RVOHSI x + v
GNU nano 8.7.1 sigstp.c
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_sigstp(int sig) {
    printf("\nSIGTSTP received\n");
    printf("Process suspension request handled\n");
}

int main() {
    signal(SIGTSTP, handle_sigstp);

    printf("Process Group ID: %d\n", getpgrp());
    printf("Program running...\n");
    printf("Press Ctrl+Z to send SIGTSTP\n");

    while (1)
        sleep(1);

    return 0;
}

```

```

sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ nano sigstp.c
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ gcc sigstp.c -o sigstp
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$ ./sigstp
Process Group ID: 1397
Program running...
Press Ctrl+Z to send SIGTSTP
^Z
SIGTSTP received
Process suspension request handled
^Z
SIGTSTP received
Process suspension request handled
^C
sai_pranavi@LAPTOP-RVOHSPKQ:~/ShellForge$

```

```
sai_pranavi@LAPTOP-RVOHSI  ×  +  ▾
GNU nano 8.7.1                q21.c *
#include <stdio.h>
#include <stdlib.h>

void module1() {
    printf("Module 1: Input processing successful\n");
}

void module2() {
    printf("Module 2: Command processing successful\n");
}

void module3() {
    printf("Module 3: Output processing successful\n");
}

int main() {
    printf("Integrated System\n");

    module1();
    module2();
    module3();

    printf("All modules executed successfully\n");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSI  ×  +  ▾
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano q21.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc q21.c -o q21
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./q21
Integrated System
Module 1: Input processing successful
Module 2: Command processing successful
Module 3: Output processing successful
All modules executed successfully
sai_pranavi@LAPTOP-RVOHSPKQ:~$ |
```

```
sai_pranavi@LAPTOP-RVOHSI  ×  +  ∨  
GNU nano 8.7.1 22.c *  
#include <stdio.h>  
#include <stdlib.h>  
  
int main() {  
    int *ptr;  
  
    ptr = malloc(5 * sizeof(int));  
  
    for (int i = 0; i < 5; i++)  
        ptr[i] = i + 1;  
  
    printf("Memory allocated successfully\n");  
  
    free(ptr);  
  
    printf("Memory released successfully\n");  
  
    return 0;  
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano 22.c  
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc -g q22.c -o q22  
cc1: fatal error: q22.c: No such file or directory  
compilation terminated.  
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc -g 22.c -o 22  
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./22  
Memory allocated successfully  
Memory released successfully
```



```
sai_pranavi@LAPTOP-RVOHSI × + v
GNU nano 8.7.1 23.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int pipe1[2];
    int pipe2[2];

    pipe(pipe1);
    pipe(pipe2);

    if (fork() == 0) {
        close(pipe1[0]);
        dup2(pipe1[1], STDOUT_FILENO);
        close(pipe1[1]);

        execlp("seq", "seq", "1", "10", NULL);
    }

    if (fork() == 0) {
        close(pipe1[1]);
        close(pipe2[0]);

        dup2(pipe1[0], STDIN_FILENO);
        dup2(pipe2[1], STDOUT_FILENO);

        close(pipe1[0]);
        close(pipe2[1]);

        execlp("grep", "grep", "1", NULL);
    }

    if (fork() == 0) {
        close(pipe1[0]);
        close(pipe1[1]);
        close(pipe2[1]);

        dup2(pipe2[0], STDIN_FILENO);
        close(pipe2[0]);

        execlp("wc", "wc", "-l", NULL);
    }

    close(pipe1[0]);
    close(pipe1[1]);
    close(pipe2[0]);
    close(pipe2[1]);

    wait(NULL);
    wait(NULL);
    wait(NULL);

    printf("Large pipeline executed successfully\n");

    return 0;
}
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano 23.c
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc 23.c -o 23
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./23
2
Large pipeline executed successfully
sai_pranavi@LAPTOP-RVOHSPKQ:~$ |
```

24

```
GNU nano 8.7.1 24.c *
#include <stdio.h>
#include <unistd.h>

void show_system() {
    printf("OSSP Project Demo\n");
    printf("-----\n");
    printf("1. Command History\n");
    printf("2. Pipe Communication\n");
    printf("3. Process Management\n");
    printf("4. Signal Handling\n");
    printf("5. Memory Management\n");
}

int main() {
    printf("Project Repository Demo\n\n");

    show_system();

    printf("\nCurrent Process ID: %d\n", getpid());
    printf("Demo completed successfully\n");

    return 0;
}
```

Memory Released Successfully

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ nano 24.c
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ gcc 24.c -o 24
```

```
sai_pranavi@LAPTOP-RVOHSPKQ:~$ ./24
```

Project Repository Demo

OSSP Project Demo

1. Command History
2. Pipe Communication
3. Process Management
4. Signal Handling
5. Memory Management

Current Process ID: 646

Demo completed successfully